SLO Handbook - FS100

Owner: SRE. Applies to: the chat, upload and login journeys of FS100. Last reviewed: 2026-10.

1. Why service level objectives

A service level objective (SLO) is a target for how reliable a user journey must be, measured over a rolling window. It turns "is the system good enough?" into a number that engineering and product agree on. Reliability above the target is not a goal in itself: every extra nine costs effort that could go into features.

2. Service level indicators

A service level indicator (SLI) is the measured ratio of good events to all events.
FS100 uses three SLIs:
- Chat availability: share of chat requests that end with a done event instead of an error event or a timeout.
- Chat latency: share of chat requests whose first token arrives within 3 seconds.
- Upload success: share of uploaded documents that reach the status ready within 5 minutes.

SLIs are measured at the NestJS API, as close to the user as possible, not inside the AI service.

3. Targets

All targets use a rolling 28-day window.
- Chat availability: 99.5 percent.
- Chat latency (first token within 3 seconds): 95 percent.
- Upload success: 99 percent.
- Login availability: inherited from Keycloak, 99.9 percent.

4. Error budget

The error budget is the allowed amount of unreliability: 100 percent minus the target. For chat availability that is 0.5 percent of all chat requests in 28 days, or roughly 3 hours and 22 minutes of full outage.

5. Error budget policy

- While budget remains, releases ship normally.
- When more than 50 percent of the budget is consumed in the first half of the window, risky changes need a second reviewer.
- When the budget is exhausted, feature releases stop. Only fixes for reliability and security ship until the SLI is back above target for 7 days.
- A single incident that consumes more than 20 percent of the budget requires a postmortem, even if it was a SEV3.

6. Alerting on burn rate

Alerts fire on the burn rate: how fast the error budget is consumed relative to the target. A burn rate of 1 uses exactly the whole budget in 28 days.
Multi-window alerts avoid both slow detection and flapping:
- Page: burn rate above 14.4 over the last hour and over the last 5 minutes (2 percent of the budget in one hour).
- Page: burn rate above 6 over the last 6 hours and over the last 30 minutes (5 percent of the budget in 6 hours).
- Ticket: burn rate above 1 over the last 3 days.
The short window makes the alert stop quickly once the problem is fixed.

7. Review cadence

SLOs are reviewed every quarter with product. Questions for the review: did users complain while the SLO was met (target too loose)? Did we burn budget on events users did not notice (SLI measures the wrong thing)? Changes to targets are recorded in the changelog of this handbook.